

ASSIGNMENT #12 November 22, 2022

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SECTION 9.2

Exercise 1 (9.2.2 — Noah Palmer). *Let $\mathbb{F}$ be a finite field of order q and let $f(x)$ be a polynomial in $\mathbb{F}[x]$ of degree $n \geq 1$. Prove that $\mathbb{F}[x]/(f(x))$ has q^n elements. [Use the preceding exercise.]*

Proof. Let $\mathbb{F}$ be a finite field of order q and let $f(x)$ be a polynomial in $\mathbb{F}[x]$ of degree $n \geq 1$. From the previous exercise we know that the elements $\bar{1}, \bar{x}, \dots, \overline{x^{n-1}}$ form a basis for the vector space $\mathbb{F}[x]/(f(x))$. Thus, each element of $\mathbb{F}[x]/(f(x))$ has the form $g(x) = a_0\bar{1} + a_1\bar{x} + \dots + a_{n-1}\overline{x^{n-1}}$ for $a_i \in \mathbb{F}$. Since there are q elements in $\mathbb{F}$ we have q options for each a_i . There are q^n choices for the n coefficients of each element in $\mathbb{F}[x]/(f(x))$ and so there are q^n possible elements in $\mathbb{F}[x]/(f(x))$. $\square$

Exercise 2 (9.2.3 — Oleksandr Bobrovnikov). *Let $f(x)$ be a polynomial in $F[x]$. Prove that $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible. [Use Proposition 7, Section 8.2.]*

Proof. We will use the following facts:

- (i) If F is a field then $F[x]$ is a PID.
- (ii) In a PID every nonzero prime ideal is maximal.
- (iii) Let R be a commutative ring. The ideal $M \subset R$ is maximal if and only if R/M is a field.
- (iv) Let R be a PID. Then $r \in R$ is prime if and only if it is irreducible.
- (v) Every maximal ideal of a commutative ring is a prime ideal.

Suppose $f(x)$ is irreducible. Then $f(x)$ is prime. Since $\deg f(x) \geq 1$, we deduce that $(f(x))$ is prime and nonzero, and thus is maximal. Since $F[x]$ is a commutative ring, we deduce that $F[x]/(f(x))$ is a field.

Suppose now $F[x]/(f(x))$ is a field. Then $(f(x))$ is maximal, and thus is prime and nonzero, so $f(x)$ is prime. Since $F[x]$ is a PID, $f(x)$ is irreducible. $\square$

Exercise 3 (9.2.7 — Glen Woodworth). *Determine all the ideals of the ring $\mathbb{Z}[x]/(2, x^3+1)$.*

Proof. By the correspondence isomorphism theorem for rings, we know that the ideals of $\mathbb{Z}[x]/(2, x^3+1)$ will bijectively correspond with the ideals of $\mathbb{Z}[x]$ containing $(2, x^3+1)$. Since 2 is irreducible in $\mathbb{Z}[x]$, we know that 2 must be one of the generators in any ideal containing (2). Note that $x^3+1 = (x+1)(x^2-x+1)$ so that $(2, x+1)$ and $(2, x^2-x+1)$ are both ideals in $\mathbb{Z}[x]$ containing $(2, x^3+1)$. Moreover, $x+1$ and x^2-x+1 are both irreducible in $\mathbb{Z}[x]$. Thus, there are only two proper ideals in $\mathbb{Z}[x]/(2, x^3+1)$. Thus, $(2, x+1)/(2, x^3+1)$ and $(2, x^2-x+1)/(2, x^3+1)$ are the only unique proper ideals in $\mathbb{Z}[x]/(2, x^3+1)$. $\square$

SECTION 9.3

Exercise 4 (9.3.1 — Glen Woodworth). *Let R be an integral domain with quotient field $\mathbb{F}$ and let $p(x)$ be a monic polynomial in $R[x]$. Assume that $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $\mathbb{F}[x]$ of smaller degree than $p(x)$. Prove that if $a(x) \notin R[x]$ then R is not a unique factorization domain. Deduce that $\mathbb{Z}[2\sqrt{2}]$ is not a UFD.*

Proof. Suppose $a(x) \notin R[x]$. For a contradiction, suppose R is a UFD. Note that $a(x) \neq 1$ because a is monic and $a(x) \notin R[x]$. If $b(x)$ is constant, then $b(x) = 1$, so $a(x) = p(x) \in R[x]$, a contradiction. Now, suppose $a(x), b(x)$ are nonconstant polynomials. By Gauss' Lemma there are nonzero elements $r, s \in \mathbb{F}$ such that $ra(x) = a'(x)$ and $sb(x) = b'(x)$ both lie in $R[x]$ and $p(x) = a'(x)b'(x)$ is a factorization in $R[x]$. We have

$$p(x) = a'(x)b'(x) = ra(x)sb(x) = rsa(x)b(x).$$

Since p, a , and b are all monic polynomials, and $a(x)b(x)$ is monic, which means $rs = 1 \in R$. Note that since $a(x)$ is monic while $ra(x) = a'(x) \in R[x]$, we know $r \in R[x]$. Similarly, since $b(x)$ is monic and $sb(x) = b'(x) \in R[x]$, s must be in $R[x]$ as well. Then $a(x) = 1a(x) = sra(x) \in R[x]$, a contradiction because we assumed $a(x) \notin R[x]$.

Consider $\mathbb{Z}[2\sqrt{2}]$. We have $x^2 - 2 \in \mathbb{Z}[2\sqrt{2}][x]$. Note $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$. Since $1/2 \in \mathbb{Q}[2\sqrt{2}]$, we know $\pm\sqrt{2} \in \mathbb{Q}[2\sqrt{2}]$ so that $(x - \sqrt{2}), (x + \sqrt{2}) \in \mathbb{Q}[2\sqrt{2}][x]$. Since $\sqrt{2} \notin \mathbb{Z}[2\sqrt{2}]$, we know $(x - \sqrt{2}), (x + \sqrt{2}) \notin \mathbb{Z}[2\sqrt{2}][x]$. By the result above, $\mathbb{Z}[2\sqrt{2}]$ is not a UFD. $\square$

Exercise 5 (9.3.3 — Noah Palmer). *Let $\mathbb{F}$ be a field. Prove that the set R of polynomials in $\mathbb{F}[x]$ whose coefficient of x is equal to 0 is a subring of $\mathbb{F}[x]$ and that R is not a UFD. [Show that $x^6 = (x^2)^3 = (x^3)^2$ gives two distinct factorization of x^6 into irreducibles.]*

Proof. Let $\mathbb{F}$ be a field and consider the set R of polynomials in $\mathbb{F}[x]$ whose coefficient of x is equal to 0. Observe that R is nonempty since $1 \in R$. Now let $p(x), q(x) \in R$ such that $p(x) = p_0 + p_2x^2 + \dots + p_kx^k$ and $q(x) = q_0 + q_2x^2 + \dots + q_nx^n$ and assume without loss of generality that $k < n$. Observe that $p(x) - q(x) = (p_0 - q_0) + (p_2 - q_2)x^2 + \dots + (p_k - q_k)x^k + \dots - q_nx^n \in R$ and $p(x)q(x) = p_0q_0 + (p_0q_2 + q_0p_2)x^2 + \dots + p_kq_nx^{n+k} \in R$. Hence, R is a subring of $\mathbb{F}[x]$. Note that x^2 and x^3 are irreducible in R since $x \notin R$. Hence, $x^6 = (x^2)(x^2)(x^2)$ and $x^6 = (x^3)(x^3)$ and so R is not a UFD. $\square$

SECTION 9.4

Exercise 6 (9.4.1 — Noah Palmer). *Determine whether the following polynomials are irreducible in the rings indicated. For those that are reducible, determine their factorization into irreducibles. The notation $\mathbb{F}_p$ denotes the finite field $\mathbb{Z}/p\mathbb{Z}$, p a prime.*

(a) $x^2 + x + 1$ in $\mathbb{F}_2[x]$.

(b) $x^3 + x + 1$ in $\mathbb{F}_3[x]$.

(c) $x^4 + 1$ in $\mathbb{F}_5[x]$.

(d) $x^4 + 10x^2 + 1$ in $\mathbb{Z}[x]$.

Proof. We show that the following polynomials are either irreducible or reducible.

- (a) Note that $x^2 + x + 1$ has no roots in $\mathbb{F}_2$ and so by Proposition 10 in Section 9.4 it is irreducible.
- (b) Note that $x^3 + x + 1$ has the root $x = 1 \pmod{3}$ in $\mathbb{F}_3$ and so it is reducible. We can factor it as follows $x^3 + x + 1 = (x + 2)(x^2 + x + 2)$.
- (c) Note that $x^4 + 1 \equiv x^4 - 4 = (x^2 - 2)(x^2 + 2)$ thus $x^4 + 1$ is reducible in $\mathbb{F}_5[x]$.
- (d) We apply the rational roots test, note that 1 is the only integer which divides a_0 and is the only integer which divides a_4 . Thus, the only possible roots of $x^4 + 10x^2 + 1$ are 1 and -1 . Since $(1)^4 + 10(1)^2 + 1 \neq 0$ and $(-1)^4 + 10(-1)^2 + 1 \neq 0$ it follows that it has no roots. Now to check if $x^4 + 10x^2 + 1$ has any quadratic factors we suppose that $(x^2 + ax + b)(x^2 + cx + d) = x^4 + 10x^2 + 1$. Thus, $bd = 1$, $(ac + b + d)x^2 = 10x^2$, $(a + c)x^3 = 0$, and $(bc + ad)x = 0$. Hence, $a = -c$ and $b, d = 1$ or -1 . If $b, d = 1$ then $2 - c^2 = 10$ which has no integer solutions. Instead, if $b, d = -1$ then $-2 - c^2 = 10$ which has no integer solutions and so no quadratic factors exist. Thus, since no linear or quadratic factors exist it follows that $x^4 + 10x^2 + 1$ is irreducible.

□

Exercise 7 (9.4.2 — Oleksandr Bobrovnikov). *Prove that the following polynomials are irreducible in $\mathbb{Z}[x]$:*

(a) $x^4 - 4x^3 + 6$

Proof. Using Eisenstein's Criterion for $2 \in \mathbb{P}$, since $2 \mid 4$, $2 \mid 6$, and $2^2 \nmid 6$, we deduce that this polynomial is irreducible over $\mathbb{Z}$. □

(b) $x^6 + 30x^5 - 15x^3 + 6x - 120$

Proof. Using Eisenstein's Criterion for $3 \in \mathbb{P}$, since $3 \mid 30$, $3 \mid 15$, $3 \mid 6$, $3 \mid 120$ and $3^2 \nmid 12$, we deduce that this polynomial is irreducible over $\mathbb{Z}$. □

(c) $x^4 + 4x^3 + 6x^2 + 2x + 1$

Proof. Let $x = t - 1$. Then $x^4 + 4x^3 + 6x^2 + 2x + 1 = (t - 1)^4 + 4(t - 1)^3 + 6(t - 1)^2 + 2(t - 1) + 1 = t^4 - 2t + 2$. Using Eisenstein's Criterion for $2 \in \mathbb{P}$, since $2 \mid 2$, and $2^2 \nmid 2$, we deduce that this polynomial is irreducible over $\mathbb{Z}$. $\square$

(d) $\frac{(x+2)^p - 2^p}{x}$, where p is an odd prime.

Proof. Note that

$$(1) \quad \frac{(x+2)^p - 2^p}{x} = \frac{\sum_{k=0}^p \binom{p}{k} x^{p-k} 2^k - 2^p}{x} = \frac{\sum_{k=0}^{p-1} \binom{p}{k} x^{p-k} 2^k}{x} = \sum_{k=0}^{p-1} \binom{p}{k} x^{p-1-k} 2^k =$$

$$x^{p-1} + \underbrace{2px^{p-2} + \cdots + 2^{p-3}(p-1)px + 2^{p-1}p}_{p \text{ divides all coefficients}}.$$

Using Eisenstein's Criterion for $p \in \mathbb{P}$, since $p^2 \nmid 2^{p-1}p$, we deduce that this polynomial is irreducible over $\mathbb{Z}$. $\square$

Exercise 8 (9.4.5 — Glen Woodworth). Find all the monic irreducible polynomials of degree ≤ 3 in $\mathbb{F}_2[x]$, and the same in $\mathbb{F}_3[x]$.

Proof. We need only check the polynomials in $\mathbb{F}_2[x]$ containing 1 because these are the only ones that may be irreducible. We must check $1, x+1, x^2+1, x^2+x+1, x^3+1, x^3+x+1, x^3+x^2+1$, and x^3+x^2+x+1 . Clearly, 1 and $x+1$ are irreducible but $x^2+1 = (x+1)^2$ in $\mathbb{F}_2[x]$, so x^2+1 is reducible. We now apply Proposition 10, which states that a polynomial of degree two or three is reducible over a field F if and only if it has a root in F . For $a(x) = x^2+x+1$, we have $a(0) = 1$ and $a(1) = 1$, so $a(x)$ has no roots in $\mathbb{F}_2$. Thus, $a(x) = x^2+x+1$ is irreducible. For $b(x) = x^3+1$, we have $b(1) = 0$, so $b(x) = x^3+1$ is reducible. For $c(x) = x^3+x+1$, we have $c(0) = 1$ and $c(1) = 1$. Hence, $c(x) = x^3+x+1$ is irreducible. For $d(x) = x^3+x^2+1$, we have $d(0) = 1$ and $d(1) = 1$, so $d(x) = x^3+x^2+1$ is irreducible. For $e(x) = x^3+x^2+x+1$, note $e(1) = 0$, so $e(x) = x^3+x^2+x+1$ is reducible. Hence, the monic irreducible polynomials of degree ≤ 3 in $\mathbb{F}_2[x]$ are $1, x+1, x^2+x+1, x^3+x+1$, and x^3+x^2+1 .

For $\mathbb{F}_3[x]$, we must check polynomials of the following forms $x+a, x^2+ax+b$, and x^3+

$ax^2 + bx + c$ where $a, b, c \in \mathbb{F}_3$. For $x + a$, we have to check $x + 1$, $x + 2$, $2x + 1$, and $2x + 2$. Note that $2x + 2 = 2(x + 1)$ where 2 is not a unit in $\mathbb{F}[x]$, so $2x + 2$ is reducible. On the other hand, it is clear that $x + 1$, $x + 2$, and $2x + 1$ are irreducible because no constant can be factored out of any of these in the integers. For $x^2 + ax + b$, we apply Proposition 10. Using the quadratic formula, we have

$$x = \frac{-a \pm \sqrt{a^2 - 4b}}{2}.$$

We check the five possibilities, $a = 0$ and $b = 1$, $a = 0$ and $b = 2$, $a = 1$ and $b = 1$, $a = 1$ and $b = 2$, and $a = b = 2$. If $a = 0$ and $b = 1$, we have

$$x = \sqrt{-4}/2 \equiv \sqrt{2}/2 \pmod{3},$$

which is not an integer so that $x^2 + 1$ is irreducible by Proposition 10. If $a = 0$ and $b = 2$, we have

$$x = \sqrt{-8}/2 \equiv \sqrt{1}/2 \pmod{3},$$

which is not an integer so that $x^2 + 2$ is irreducible. If $a = 1$ and $b = 1$, we have

$$x = (-1 \pm \sqrt{1 - 4})/2 \equiv 2/2 \equiv 1 \pmod{3}.$$

To check, we have $1^2 + 1(1) + 1 = 3 \equiv 0 \pmod{3}$, so $x^2 + x + 1$ is reducible. If $a = 1$ and $b = 2$, we have

$$x = (-1 \pm \sqrt{1 - 8})/2 \equiv (2 \pm \sqrt{2})/2 \pmod{3},$$

which is not an integer so $x^2 + x + 2$ is irreducible. If $a = b = 2$, we have

$$x = (-2 \pm \sqrt{4 - 4})/2 \equiv 1/2 \pmod{3},$$

which is not an integer so $x^2 + 2x + 2$ is irreducible. We now need to check polynomials of the form $p(x) = x^3 + ax^2 + bx + c$. We check each element of $r \in \mathbb{F}_3$ to see if a , b , and c could be chosen so that $p(r) = 0$ but we omit the trivial cases. We have $p(1) = 1 + a + b + c$. So long as $a + b + c \equiv 2 \pmod{3}$, we have $p(1) \equiv 0 \pmod{3}$. Since anytime $c = 0$, $p(x)$ is

clearly reducible for any x , $a + b + c \equiv 2 \pmod{3}$ means that the following combinations of (a, b, c) will ensure $p(1)$ is reducible: $(a, b, c) = (0, 1, 1), (1, 0, 1), (0, 0, 2)$. Now, note $p(2) \equiv 2 + a + 2b + c \pmod{3}$. To ensure $p(2)$ has a root, we need $a + 2b + c \equiv 1 \pmod{3}$. Once again, we rule out $c = 0$ as a trivial case, so we need only check when $a + 2b + 1 \equiv 1 \pmod{3}$ and $a + 2b + 2 \equiv 1 \pmod{3}$. Simplifying, we need only check $a + 2b \equiv 0 \pmod{3}$ and $a + 2b \equiv 2 \pmod{3}$. Then the following combinations of (a, b, c) will ensure $p(2)$ is reducible: $(a, b, c) = (0, 0, 1), (1, 1, 1), (2, 2, 1), (0, 1, 2), (1, 2, 2), (2, 0, 2)$. Thus, every other combination of a, b , and c will be reducible (so long as $c \neq 0$). The monic irreducible polynomials of degree ≤ 3 in $\mathbb{F}_3[x]$ are $1, x+1, x+2, 2x+1, x^2+1, x^2+2, x^2+x+2, x^2+2x+2, x^3+2x+1, x^3+2x+2, x^3+x^2+x+2, x^3+x^2+2x+1, x^3+x^2+2, x^3+2x^2+1, x^3+2x^2+x+2, x^3+2x^2+x+1, x^3+2x^2+2x+2$. $\square$

Exercise 9 (9.4.6 — Stefano Fochesatto). *Construct fields for which the following orders:*
 (a) 9, (b) 49, (c) 8, and (d) 81.

(a) *A field of order 9.*

Proof. Consider $F = \mathbb{F}_3[x]$ a field of order 3. Let $f(x) = x^2 + 1$ an irreducible polynomial of order 2 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 9. $\square$

(b) *A field of order 49.*

Proof. Consider $F = \mathbb{F}_7[x]$ a field of order 7. Let $f(x) = x^2 + 1$ an irreducible polynomial of order 2 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 49. $\square$

(c) *A field of order 8.*

Proof. Consider $F = \mathbb{F}_2[x]$ a field of order 2. Let $f(x) = x^3 + x + 1$ an irreducible polynomial of order 3 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 8. $\square$

(d) *A field of order 81.*

Proof. Consider $F = \mathbb{F}_3[x]$ a field of order 3. Let $f(x) = x^4 + x + 2$, showing that $f(x)$ is irreducible in F implies that $F/(f(x))$ is order $3^4 = 81$ as we have done in previous exercises. Clearly $f(x) = x^4 + x + 2$ has no linear factors as it has no roots in $\mathbb{F}_3$, $f(0) = 1$, $f(1) = 1$, and $f(2) = 2$. Suppose for the sake of contradiction that $f(x)$ has quadratic factors

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + c)x^3 + (ac + b + d)x^2 + (ad + bc)x + bd.$$

Therefore we get that $a + c = 0$, $ac + b + d = 0$, $ad + bc = 1$, and $bd = 2$. Since $c = -a$, by substitution we get $ad + bc = a(d - b) = 1$. So $b \neq d$ and since $bd = 2$, either $b = 2$ and $d = 1$ or vice versa. Let $b = 2$ and $d = 1$ then $a(d - b) = 1$ gives that $a = -1$ and therefore $c = 1$, however we get that $-1 = (-1)(1) + 2 + 1 = ac + b + d = 0$ a contradiction. Let $b = 1$ and $d = 2$ then $a(d - b) = 1$ gives that $a = 1$ and therefore $c = -1$, however $-1 = (1)(-1) + 1 + 2 = ac + b + d = 0$ so we arrive at another contradiction. Therefore $f(x)$ is irreducible in F and thus $F/(f(x))$ is order $3^4 = 81$. $\square$

Exercise 10 (9.4.18 — Stefano Fochesatto). *Show that $6x^5 + 14x^3 - 21x + 35$ and $18x^5 - 30x^2 + 120x + 360$ are irreducible over $\mathbb{Q}[x]$.*

Proof. Applying Eisenstein's Criterion with prime $p = 7$ to show that $6x^5 + 14x^3 - 21x + 35$ is irreducible over $\mathbb{Z}[x]$ and also applying Eisenstein's Criterion with prime $p = 5$ to show that $18x^5 - 30x^2 + 120x + 360$ is irreducible over $\mathbb{Z}[x]$. By the contrapositive of Gauss's Lemma we can conclude that both polynomials are irreducible over $\mathbb{Q}[x]$. $\square$

SECTION 9.5

Exercise 11 (9.5.1 — Oleksandr Bobrovnikov). *Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Describe the nilradical of $F[x]/(f(x))$ in terms of factorization of $f(x)$.*

Solution: Note that $F[x]$ is a UFD and suppose $f(x) = f_1(x)^{n_1} f_2(x)^{n_2} \dots f_k(x)^{n_k}$, where $f_j(x)$ are irreducible. Recall that in the exercise 7.4.30 we proved that $(\text{rad } I)/I = \mathfrak{N}(R/I)$ for any commutative ring R and its ideal I . So $\mathfrak{N}(F[x]/(f(x))) = [\text{rad}(f(x))]/(f(x))$. Note that $\text{rad}(f(x)) = \{g(x) \mid g(x)^n \in (f(x)) \text{ for some } n \in \mathbb{Z}^+\}$. Then $g(x) \in \text{rad}(f(x))$ if and only if $f_j(x) \mid g(x)$ for every $1 \leq j \leq k$. So $\text{rad}(f(x)) = (f_1(x)f_2(x) \dots f_k(x))$, and thus $\mathfrak{N}(F[x]/(f(x))) = (f_1(x)f_2(x) \dots f_k(x))/(f(x))$. $\square$

Exercise 12 (9.5.3 — Stefano Fochesatto). *Let p be an odd prime in $\mathbb{Z}$ and let n be a positive integer. Prove that $x^n - p$ is irreducible over $\mathbb{Z}[i]$.*

Proof. Suppose p is an odd prime in $\mathbb{Z}$ and let n be a positive integer. Since p is an odd prime in $\mathbb{Z}$ it is either $p \cong 3 \pmod{4}$ or $p \cong 1 \pmod{4}$. By Proposition 18 (2.b) we know that if $p \cong 3 \pmod{4}$ then p is irreducible in $\mathbb{Z}[i]$. In which case (p) is a prime ideal in $\mathbb{Z}[i]$ which satisfies Eisenstein's Criterion since $p \notin (p)^2$. From Proposition 18 (2.c) we get that if $p \cong 1 \pmod{4}$ then $p = (a + bi)(a - bi)$ where $(a + bi)$ and $(a - bi)$ are irreducible in $\mathbb{Z}[i]$. In which case $(a + bi)$ is a prime ideal in $\mathbb{Z}[i]$ which satisfies Eisenstein's Criterion since $p \notin (a + bi)^2$. So $x^n - p$ is irreducible over $\mathbb{Z}[i]$. $\square$